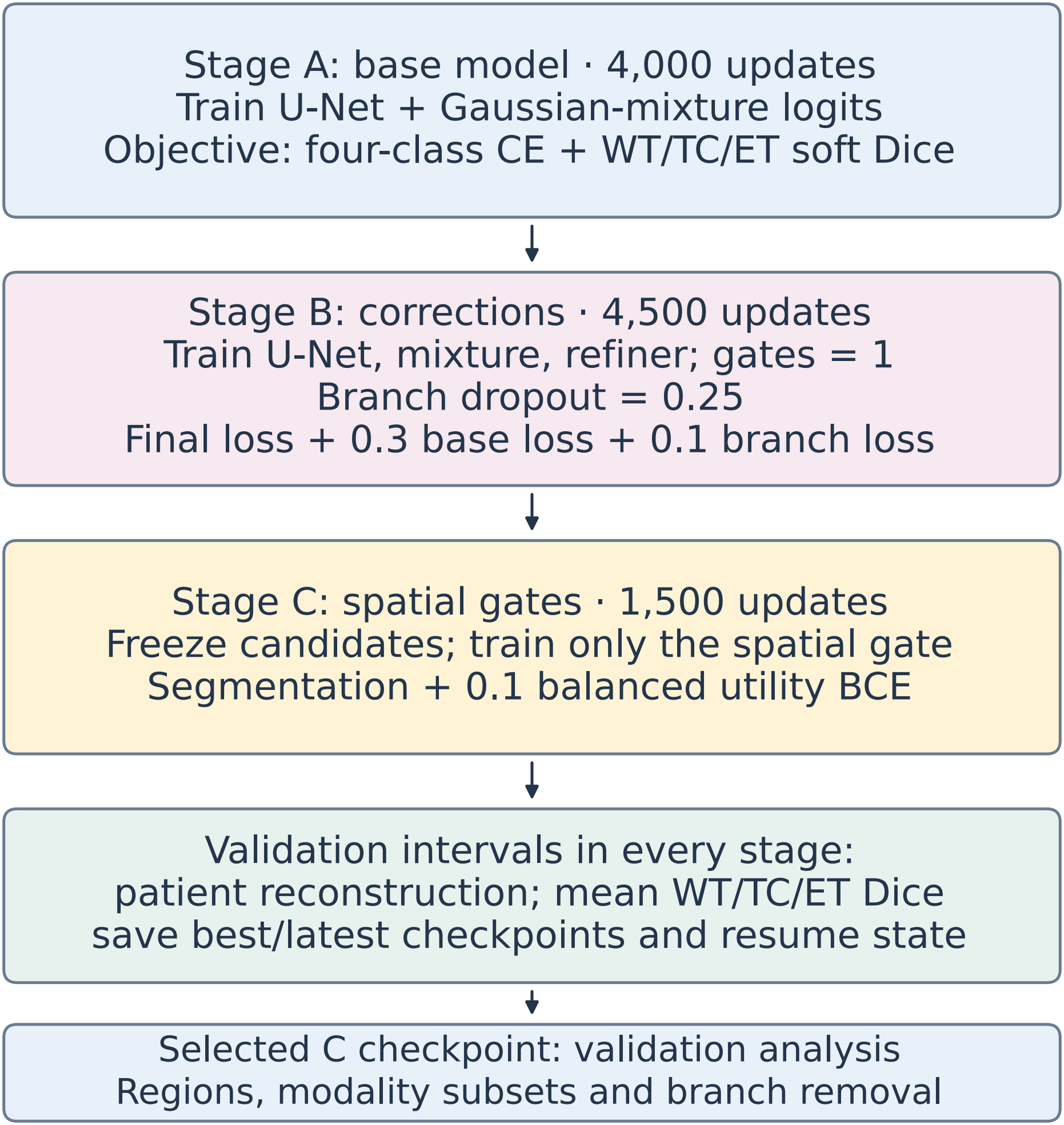
Stage A: base model · 4,000 updates
Train U-Net + Gaussian-mixture logits
Objective: four-class CE + WT/TC/ET soft Dice
Stage B: corrections · 4,500 updates
Train U-Net, mixture, refiner; gates = 1
Branch dropout = 0.25
Final loss + 0.3 base loss + 0.1 branch loss
Stage C: spatial gates · 1,500 updates
Freeze candidates; train only the spatial gate
Segmentation + 0.1 balanced utility BCE
Validation intervals in every stage:
patient reconstruction; mean WT/TC/ET Dice
save best/latest checkpoints and resume state
Selected C checkpoint: validation analysis
Regions, modality subsets and branch removal